

NEXAURA

NexClinic

Integrated Healthcare Management Platform

USER MANUAL

Operational Guide for Patients, Doctors, and Administrators

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PROJECT TEAM

M.T. Dineth
H.M.U.A. Perera
S.M.D.S.B. Samarakoon
J.G.G. Methmaka

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Abstract

NexClinic is a web based outpatient clinic, appointment, communication and healthcare administration system connecting patients, doctors, hospital administrators and system administrators. This manual documents the operational use of NexClinic by each of these roles, account registration and OTP based verification, appointment discovery and booking, medication and clinical record management, real time advice chat, hospital level doctor verification and scheduling and system level account administration. Instructions are written from direct observation of the implemented workflows and features that are simulated, limited or planned rather than complete are identified explicitly rather than presented as finished capabilities. The manual is intended to be read alongside the companion NexClinic Developer Guide, which addresses the system's internal architecture.

Index Terms

Healthcare information system, Appointment scheduling, One time password verification, Role based access control, Telehealth chat, Hospital administration, Patient portal.

I. Purpose and Scope

This manual explains how to use NexClinic, a web based outpatient clinic, appointment, communication and healthcare administration system. It is written for the four user roles supported by the system patients, doctors, hospital administrators and system administrators. Each role sees a different workspace after signing in and this manual is organized so that a reader can go directly to the section that matches their role.

The manual covers account registration, one time password (OTP) verification, login and logout, password reset, everyday patient tasks such as booking appointments and managing medication, doctor tasks such as managing appointment slots and recording clinical notes, hospital administrator tasks such as verifying doctors and generating schedules and system administrator tasks related to account and role management.

This manual follows the numbered heading, figure and table conventions associated with IEEE technical documentation [1], adapted for a university software project rather than a journal submission.

The manual does not cover source code, database design or deployment procedures. Those topics are addressed separately in the NexClinic Developer Guide, which is written for developers, testers and operators rather than end users. Where a NexClinic function is simulated or limited in its current form the appointment payment step, for example, this manual says so directly rather than describing it as a finished production feature.

II. System Overview

NexClinic's backend is built on Django and Django REST Framework [2], [3] and its interface is built with Next.js [4]. A reader who wants implementation detail on either layer should consult the Developer Guide.

NexClinic connects patients with doctors and hospitals through a single online system. A patient can search for a doctor, check available appointment times and book a consultation without phoning the clinic. A doctor can manage a personal schedule, respond to incoming bookings and record consultation outcomes. A hospital administrator can bring doctors onto the platform, build recurring schedules and monitor how the hospital's appointments and doctors are performing. A system administrator maintains the accounts and roles that the other three groups depend on.

The main functional areas of NexClinic are account and identity management, doctor discovery, appointment scheduling, clinical record keeping (medical history, prescriptions and medication tracking), messaging through an advice chat feature and a notification system that keeps every role informed of relevant events. Hospital administration adds a layer above individual doctors, hospitals verify doctors, generate slots from reusable templates and publish announcements to their patient and doctor community.

Typical Patient Journey

A new patient typically moves through NexClinic in the following order: registering an account, verifying the account by email OTP, signing in, completing a profile, searching for a suitable doctor, booking an appointment, attending the consultation and then reviewing the resulting prescription or clinical notes together with any medication reminders that were set up afterward. The journey is illustrated in Fig. 1.

Register account	Verify OTP	Sign in	Complete profile	Find a doctor	Book appointment	Attend consultation	Review records
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Fig. 1. Typical patient journey through NexClinic

Role and Responsibility Table

Table 4 summarizes the responsibilities of each role at a glance. Later sections describe each responsibility in detail with numbered instructions.

Role	Primary Responsibility	Typical Workspace
Patient	Book and manage appointments, maintain a personal health profile, track medication and communicate with a doctor through advice chat when eligible.	Patient dashboard
Doctor	Maintain a professional profile, manage appointment slots, respond to bookings and record consultation outcomes.	Doctor dashboard
Hospital Administrator	Verify and manage doctor relationships, build schedule templates, generate appointment slots and publish announcements for the hospital.	Hospital administrator dashboard
System Administrator	Maintain user accounts, roles and access and support operational troubleshooting across the platform.	System administration interface

Table 4. Role and responsibility overview

III. Getting Started

Browser and Connectivity Requirements

NexClinic is a browser based application and does not require a separate installation. A recent version of a standard desktop or mobile browser such as Chrome, Edge, Firefox or Safari is recommended, with JavaScript enabled. A stable internet connection is required throughout a session and features that depend on real time updates, such as advice chat and live notifications, require an uninterrupted connection to work correctly.

Account Requirements

Every NexClinic user needs a valid email address, because the email address is used both as the sign in identifier and as the destination for OTP codes and password reset links. Patients, doctors and hospital administrators each register through a role specific registration form. System administrator accounts are created and managed internally rather than through public self registration.

Accessing the Application

1. Open a supported browser and navigate to the NexClinic web address provided by your institution or clinic.
2. If you already have an account, select “Login”. If you do not, select “Register” and choose the account type that matches your role.
3. Follow the on-screen prompts. Fields marked as required must be completed before the form can be submitted.
4. After a successful sign in, NexClinic opens the dashboard that matches your role automatically.

Interface Navigation Conventions

NexClinic uses a consistent visual language across every workspace so that a user who has learned one section can recognize similar controls elsewhere.

- Buttons: primary actions appear as solid rectangular buttons, secondary actions appear as outlined buttons.
- Forms: required fields are marked and an inline message appears beneath a field when its value is invalid.
- Alerts: confirmation and error messages appear as a banner near the top of the page or as a small pop up notice and remain visible until dismissed or replaced.
- Tables and lists: appointment lists, doctor directories and similar collections are shown as scrollable lists or tables with column headings that describe the content.
- Filters: search and filter controls sit above a list and narrow its contents without reloading the whole page.
- Status labels: short labels such as “Pending”, “Accepted” or “Completed” indicate the current state of an appointment or request at a glance.

These conventions are chosen with general web accessibility guidance in mind, including labeled form fields and clearly distinguishable status indicators [7].

IV. Account Registration and Security

Patient Registration

1. From the landing page, select “Register” and choose “Patient”.
2. Enter your full name, email address, a password and any other requested basic details.
3. Choose a password that is difficult to guess and not reused from another account. NexClinic checks the password against basic strength requirements before accepting the form.
4. Submit the form. NexClinic creates an inactive account and sends a one-time password (OTP) to the email address you entered.
5. Proceed to the OTP verification step described below to activate the account.

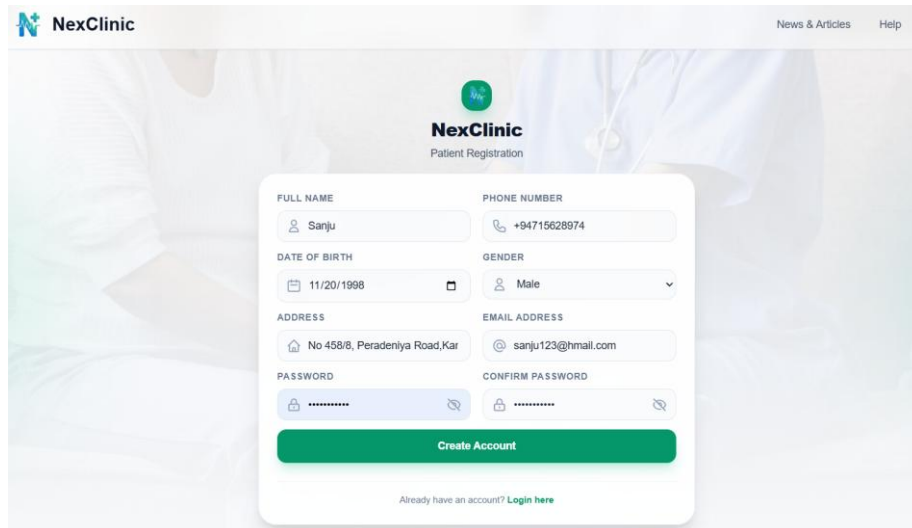
The image shows the NexClinic Patient Registration form. At the top, there is a NexClinic logo and navigation links for 'News & Articles' and 'Help'. The form is titled 'NexClinic Patient Registration'. It contains several input fields: 'FULL NAME' (with a user icon and the text 'Sanju'), 'PHONE NUMBER' (with a phone icon and the text '+94715628974'), 'DATE OF BIRTH' (with a calendar icon and the text '11/20/1998'), 'GENDER' (with a dropdown menu showing 'Male'), 'ADDRESS' (with a house icon and the text 'No 458/B, Peradeniya Road, Kar'), 'EMAIL ADDRESS' (with an email icon and the text 'sanju123@gmail.com'), 'PASSWORD', and 'CONFIRM PASSWORD'. There are eye icons to toggle password visibility. A green 'Create Account' button is at the bottom of the form. Below the button, there is a link: 'Already have an account? [Login here](#)'.

Fig. 2. Patient registration screen

Doctor Registration

Doctor registration follows the same overall pattern as patient registration but collects professional information in addition to basic identity details, such as specialization and professional registration information. A doctor account created through self registration typically remains in an unverified state with respect to hospital association until a hospital administrator reviews and approves the doctor's request to join a specific hospital, described in Section VI.

1. From the landing page, select “Register” and choose “Doctor”.
2. Enter your full name, email address, password, specialization and professional registration details as requested by the form.
3. Submit the form and complete OTP verification as described below.
4. After verifying the account, request association with one or more hospitals from the doctor dashboard so that patients can find and book you.

Hospital Administrator Registration

Hospital administrator registration is used to bring a new hospital or clinic onto NexClinic or to add an additional administrator to an existing hospital record. The form collects the administrator's identity information together with the hospital's basic profile and contact details. As with the other roles, the account must be verified by OTP before it becomes active.

Registration OTP Verification

After submitting a registration form, NexClinic sends a numeric OTP to the email address provided. The OTP verification screen asks you to enter this code to activate the account.

1. Check the inbox of the email address you registered with. The message normally arrives within a short time.
2. Enter the OTP exactly as received into the verification screen.
3. Submit the code. If it is correct and has not expired, the account becomes active and you are directed to sign in.

4. If the code is rejected, confirm that it was typed correctly and that it has not expired, then try again or request a new code.

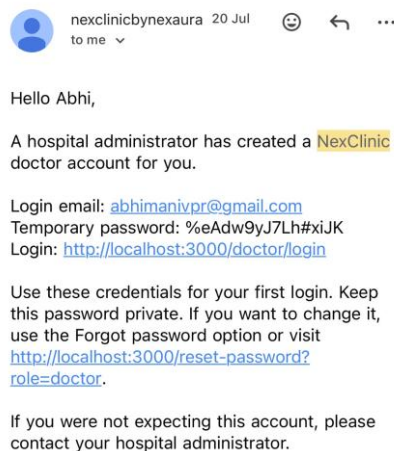


Fig. 3. Registration OTP verification screen

Resending an OTP

If the OTP email does not arrive, NexClinic provides a “Resend Code” control on the verification screen. Selecting it issues a new OTP and invalidates the previous one. To prevent abuse, the resend control may be temporarily disabled for a short period after each use if this happens, wait for the indicated interval before trying again.

OTP Expiry

Every OTP is valid for a limited time. If the code is not submitted before it expires, verification fails even if the digits are correct, and a new code must be requested. This time limit reduces the risk that an intercepted or delayed message could be used later by someone other than the intended recipient.

Registration Errors

Common registration errors include an email address that is already registered, a password that does not meet the minimum strength requirement and required fields left blank. NexClinic displays an inline message describing the specific problem so that it can be corrected without reentering the whole form.

Login

1. Open the login page and enter your registered email address and password.
2. Submit the form. NexClinic checks the credentials against the stored account.
3. If the credentials are correct, NexClinic proceeds to login OTP verification when this additional step is required for your account or session.
4. If the credentials are incorrect, an error message appears and repeated failed attempts may result in a short temporary lockout, described under Security Practices.

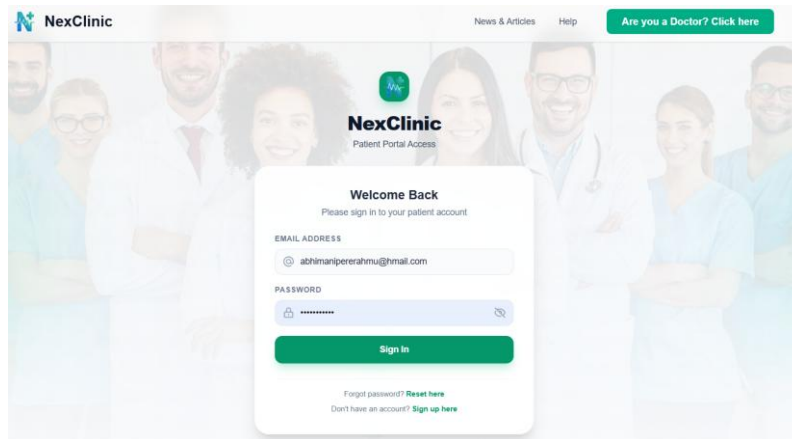


Fig. 4. Patient Login screen

Login OTP Verification

For added protection, NexClinic can require a second OTP step at login, sent to the account's registered email address. This works the same way as registration OTP verification, enter the code before it expires and use the resend control if the message does not arrive.

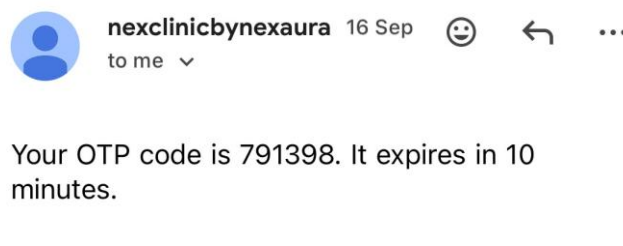


Fig. 5. Login OTP verification screen

Password Reset

1. From the login page, select "Forgot Password".
2. Enter the email address associated with your account and submit the request.
3. NexClinic sends a secure password reset link to that email address if an account exists for it.
4. Open the link, which leads to a form for entering a new password.
5. Enter and confirm the new password, meeting the same strength requirements as at registration and submit the form.
6. Sign in using the new password.

This reset workflow follows the general pattern recommended for self service password recovery, in which a time limited, single use link is sent to a verified address rather than the original password being disclosed to anyone [6].

Reset Password

Enter your new password and confirm it to reset your password.

Email Address

Send Reset Link

Fig. 6. Password reset screen

Logout

Selecting “Logout” from the account menu, available in every workspace, ends the current session and clears the credentials stored in the browser for that session. Always log out when using a shared or public computer.

Secure Account Practices

- Use a strong, unique password for NexClinic that is not reused on other websites [5].
- Never share your password, OTP codes or password reset links with anyone else, including hospital staff or people claiming to represent NexClinic.
- Sign in only from devices you trust and avoid saving your password on a shared or public computer.
- Always sign out when you finish a session on a shared device.
- Review your profile information periodically and correct anything that is no longer accurate.
- Report unexpected account activity, such as an OTP you did not request, by changing your password and contacting your hospital or system administrator.

V. Patient Operations

Patient Dashboard

After signing in, a patient is taken to the patient dashboard, which brings together the information a patient is most likely to need in a single view: upcoming appointments, recent notifications and quick links to booking a new appointment, viewing medical history and managing medication.

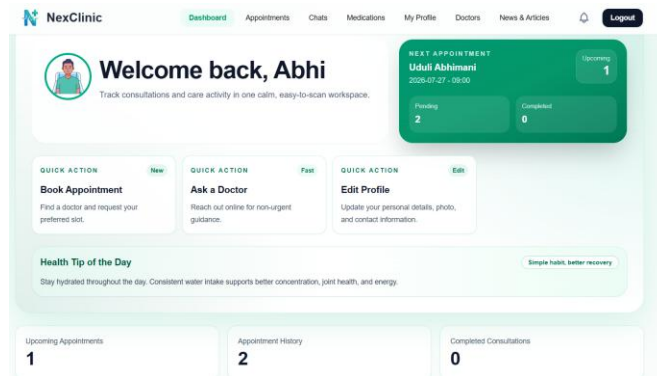


Fig. 7. Patient dashboard

Patient Profile

The patient profile page stores the personal information that doctors and hospital staff rely on when providing care. Keeping it accurate and current is one of the more important routine tasks a patient can perform in NexClinic.

Editing Personal Details

1. Open “Profile” from the patient dashboard or account menu.
2. Select “Edit” and update the relevant fields, such as contact number, address or date of birth.
3. Save the changes. A confirmation message appears once the update has been applied.

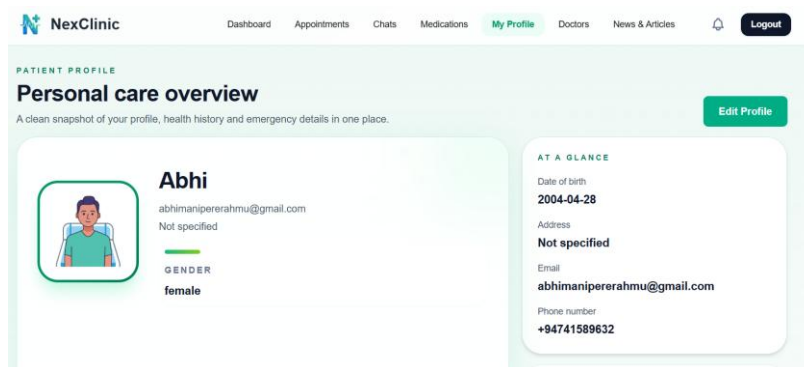


Fig. 8. Patient profile screen

Emergency Contact Information

The profile includes a section for at least one emergency contact, consisting of a name, relationship and phone number. This information is intended for use in situations where a patient cannot be reached directly, and should be kept current, particularly after a change of household or family circumstances.

Allergy Information

Patients can record known allergies in their profile. This information is intended to support safer prescribing and consultation by making relevant history visible to an authorized doctor. It should be reviewed and updated whenever a new allergy is identified or a previous entry is no longer accurate.

Medical History

The medical history section presents the clinical information that has accumulated through the patient's use of NexClinic, including past diagnoses, consultation notes made available by doctors and prescriptions. This view is read oriented from the patient's side clinical entries are created by doctors during or after a consultation as described in Section VI.

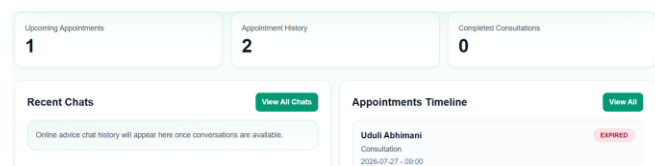


Fig. 9. Patient medical history screen

Medical Document Uploads

Patients can upload supporting medical documents, such as prior test results or referral letters, so that a doctor has relevant background before or during a consultation.

1. Open the medical documents section of the profile or medical history area.
2. Select “Upload” and choose a supported file from your device.
3. Confirm the upload. The document appears in the list once processing completes.

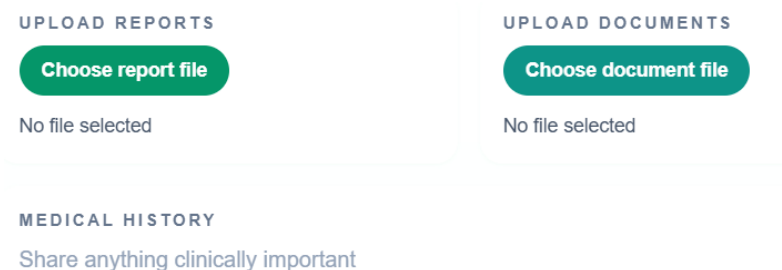


Fig. 10. Medical document upload screen

Finding a Doctor

The doctor directory lets a patient browse and search for doctors before booking. Search can be combined with filters to narrow a long list down to a manageable set of relevant options.

Filtering Doctors

- By specialization to find doctors practicing in a particular field.
- By hospital to find doctors associated with a specific hospital or clinic location.

- By name, using the search box, when the patient already knows which doctor they want to see.

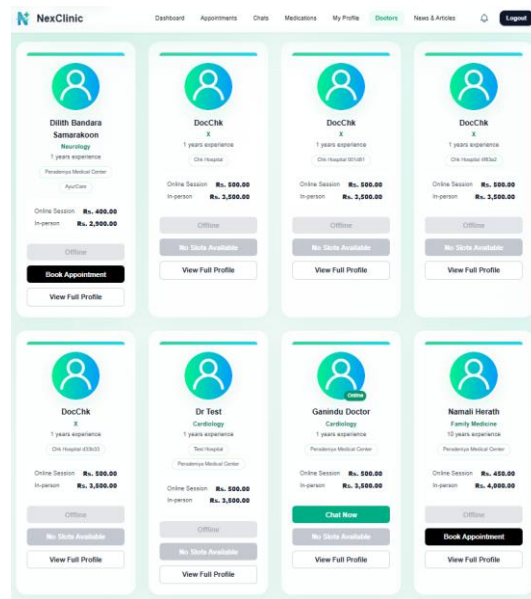


Fig. 11. Doctor directory screen

Viewing Doctor Details

Selecting a doctor from the directory opens a profile page showing qualifications, years of experience, specialization, the hospitals the doctor is associated with and the consultation fee. This information is intended to help a patient decide whether the doctor is a good match before booking.

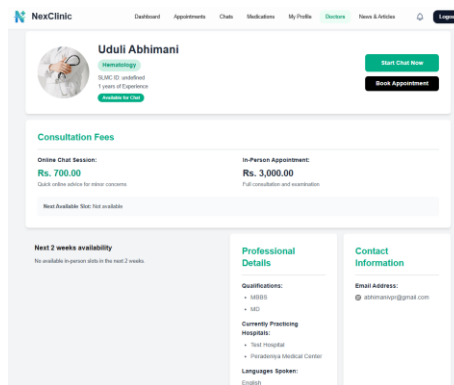


Fig. 12. Doctor profile screen

Viewing Hospital Information

Where a doctor is associated with more than one hospital, the doctor profile lists each hospital together with the associated consultation fee and available slots, so the patient can choose the location that is most convenient.

Booking an Appointment

Appointment booking in NexClinic follows a consistent sequence. Select a doctor, choose an available slot, review the details, confirm payment and submit.

1. From the doctor's profile, select "Book Appointment".

2. Choose a hospital location if the doctor is associated with more than one.
3. Select an available date and time slot from the calendar or list of open slots.
 4. Review the summary shown on screen, including the doctor's name, hospital, date, time and consultation fee.
5. Complete the payment confirmation step (see below).
6. Submit the booking. The new appointment appears in your appointment list with a status of "Pending."

The screenshot shows the 'Book an Appointment' form on the NexClinic website. The form has a light blue header with the NexClinic logo and navigation links. The main form area is white with a light blue border. It contains the following sections:

- Hospital:** A dropdown menu with the placeholder text 'Search hospitals...'.
- Doctor:** A dropdown menu with the placeholder text 'Search doctors...'.
- Available Time Slots:** A section with the text 'Please select a hospital and doctor to view available times.' and a placeholder for a calendar or list of slots.
- Reason for Visit (Optional):** A text area with the placeholder text 'Describe your symptoms...'.
- Proceed to Payment:** A green button at the bottom of the form.

Fig. 13. Appointment booking screen

This screenshot shows the 'Book an Appointment' form with specific selections. The 'Hospital' dropdown is set to 'Peradeniya Medical Center' and the 'Doctor' dropdown is set to 'Namali Herath'. Below the doctor selection, there is a small profile card for Namali Herath, marked as 'Available Now'. The 'Available Time Slots' section shows a calendar for 'MON, SEP 29, 2020' with a green button for '9:00 AM' selected. The 'Reason for Visit (Optional)' field is still present. At the bottom, there is a 'Summary' section that displays the selected doctor, location, and date & time. A green 'Proceed to Payment' button is at the bottom.

Summary	
Doctor	Namali Herath
Location	Peradeniya Medical Center
Date & Time	Mon, Sep 28, 2020 at 9:00 AM

Fig. 14. Date and slot selection screen

Reviewing Fees

The consultation fee for the selected doctor and hospital is displayed clearly before the booking is confirmed, so that there are no surprises at the payment step.

During booking, the patient is shown a payment confirmation screen summarizing the fee. Selecting "Confirm Payment" completes the mock transaction and allows the booking to proceed, no card or bank details are processed by this step.

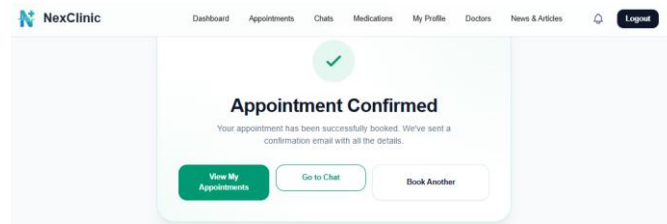


Fig. 15. Booking confirmation screen

Reviewing Appointments

The appointments list, reachable from the patient dashboard, shows all current and past appointments together with their status. Selecting an entry opens its details, including the doctor, hospital, date, time and any notes recorded so far.

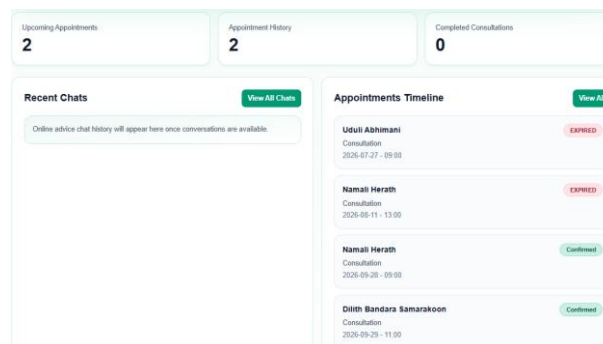


Fig. 16. Appointment list screen

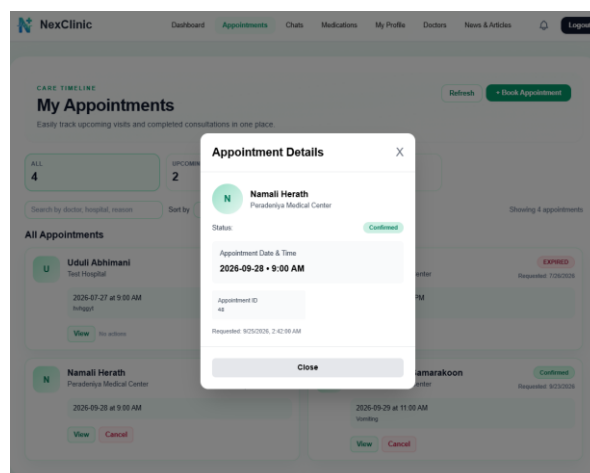


Fig. 17. Appointment details screen

Appointment Status Meanings

Table 5 explains what each appointment status means from the patient's point of view. A fuller explanation of the workflow, including which role can act at each stage, appears in Section III of the appointment related material and in the Developer Guide.

Status	Meaning for the Patient
Pending	The booking has been submitted and is waiting for the doctor to accept, reject, or propose a new time.
Accepted	The doctor has confirmed the appointment. The patient should attend at the scheduled date and time.
Reschedule Pending	The doctor has proposed a different date or time. The patient needs to review and respond to the proposal.
Completed	The consultation has taken place and the doctor has closed out the appointment. Related clinical records may now be available.
Cancelled	The appointment will not take place. It may have been cancelled by the patient or, in limited cases by the hospital.
Rejected	The doctor was unable to accept the original booking and did not propose an alternative time.

Table 5. Appointment status meanings for patients

Cancelling an Appointment

1. Open the appointment from the appointment list.
2. Select “Cancel” if the option is available for the appointment's current status.
3. Confirm the cancellation when prompted. The status changes to “Cancelled”.

Cancellation is normally available while an appointment is Pending or Accepted and is no longer offered once an appointment has moved to Completed. If “Cancel” does not appear, consult the Troubleshooting section.

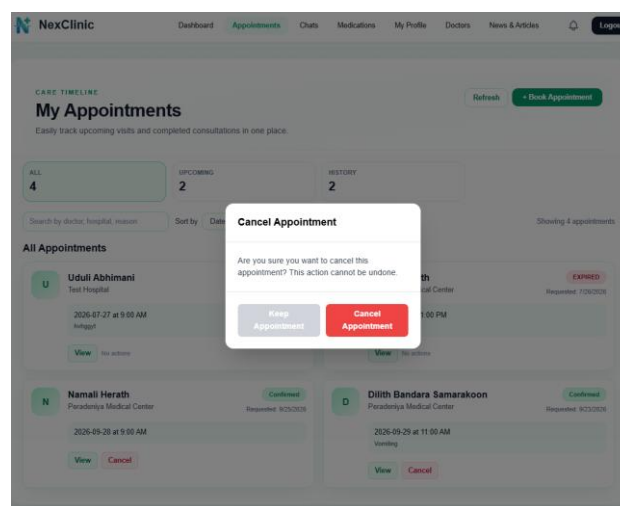


Fig. 18. Appointment status controls

Reviewing Prescriptions

Prescriptions created by a doctor during or after a consultation appear alongside the related appointment and in the medical history view, so a patient can refer to dosage and instructions without needing a paper copy.

Medication Management

Patients can maintain a personal list of medications they are taking, whether prescribed through NexClinic or elsewhere, to keep a complete picture of current treatment.

1. Open “Medications” from the patient dashboard.
2. Select “Add Medication” and enter the medication name, dosage and schedule.
3. Save the entry. It appears in the medication list, where it can be edited or removed later.

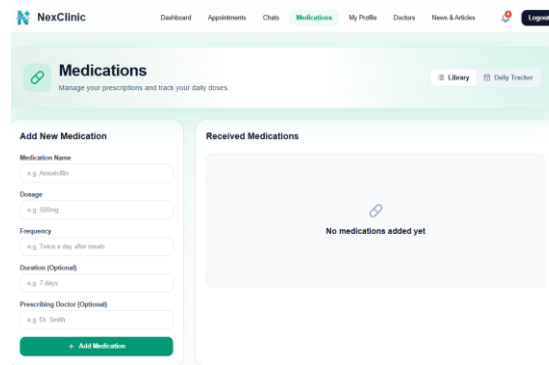


Fig. 19. Medication management screen

Medication Reminders

1. Open a medication entry and select “Add Reminder”.
2. Choose the times the reminder should trigger, matching the medication’s dosing schedule.
3. Save the reminder. NexClinic will surface it through the notification system at the configured times.

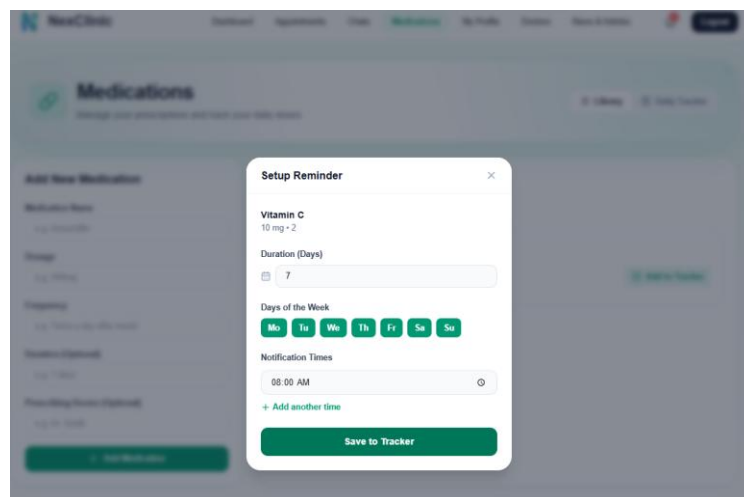


Fig. 20. Medication reminder configuration screen

Medication Logs

Each time a reminder is acted upon, or a dose is taken and logged manually, NexClinic records the event in a medication log. This log gives the patient and where authorized the treating doctor, a history of adherence over time.

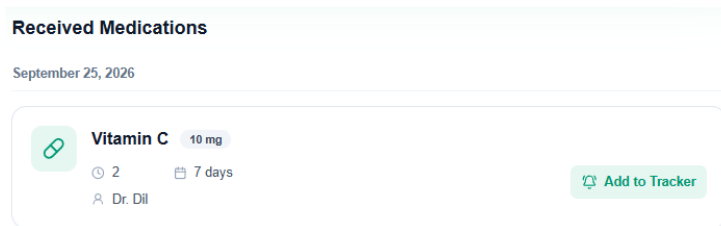


Fig. 21. Medication logs screen

Notifications

NexClinic keeps patients informed of appointment updates, reminders and hospital announcements through a notification centre reachable from every workspace.

1. Select the notification icon to open the notification centre.
2. Review the list of notifications, most recent first.
3. Select a notification to open the related item, such as an appointment or announcement.
4. Mark individual notifications as read, or use “Mark All as Read” to clear the list.

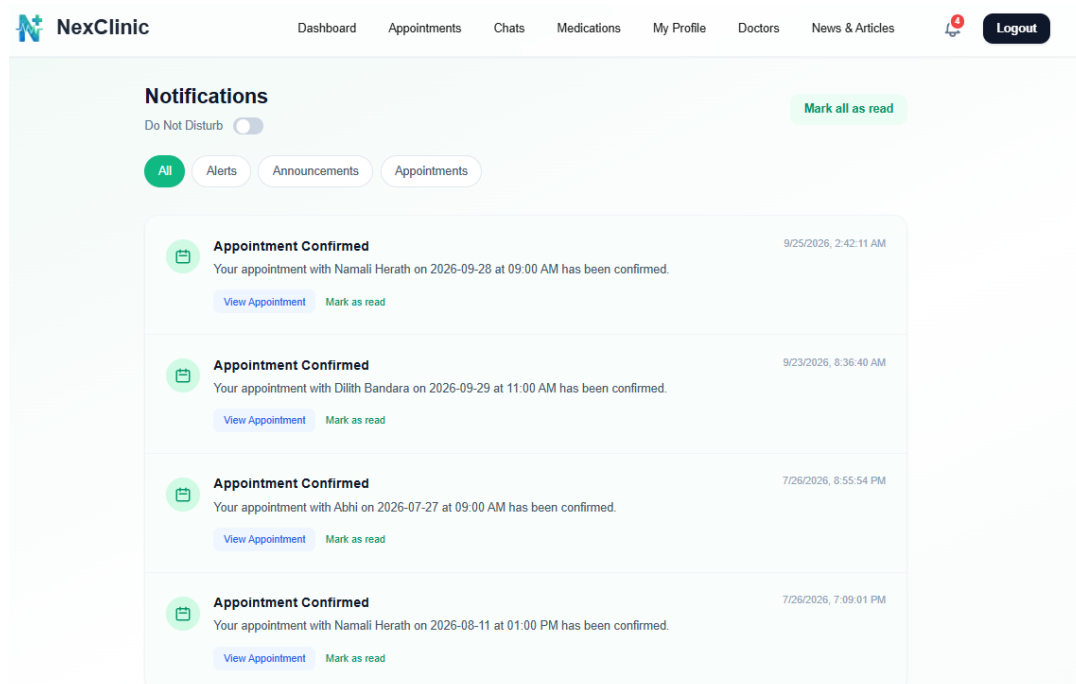


Fig. 22. Notification centre screen

Advice Chat

Advice chat allows a patient to exchange messages with a doctor outside the structure of a scheduled appointment, when the doctor has made this feature available and the patient is eligible to use it, typically in connection with an existing or recent appointment relationship.

1. Open “Advice Chat” from the patient dashboard.
2. Select an eligible doctor or existing conversation thread.

3. Type a message and select “Send”. Messages appear in the thread in real time while both participants are connected.
4. Attach a supported file, such as an image, when a written description is not sufficient.

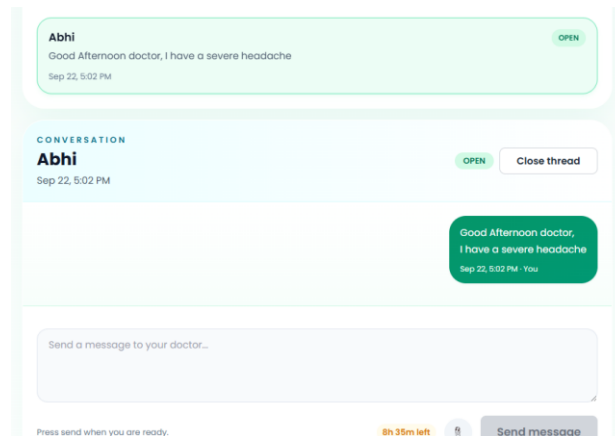


Fig. 23. Advice chat workspace screen

Chat Attachments

Supported file types can be attached directly within a chat thread. As with medical document uploads, attachments are subject to file type and size restrictions.

Closing or Reviewing Chat History

A doctor may close a chat thread once the matter has been addressed. A closed thread remains visible for reference but no longer accepts new messages. Previous eligible conversations can be reopened for viewing from the advice chat workspace at any time.

VI. Doctor Operations

Doctor Dashboard

The doctor dashboard summarizes the day's schedule, incoming appointment requests awaiting a decision and recent notifications, giving a doctor a starting point for each session in NexClinic.

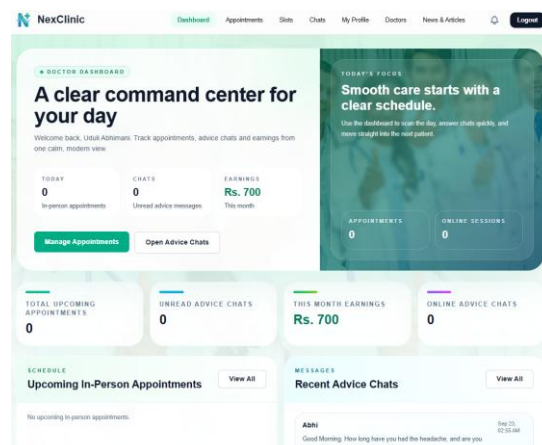


Fig. 24. Doctor dashboard screen

Doctor Profile

A doctor's profile presents the information patients see when browsing the directory and deciding whether to book. Keeping it complete and accurate directly affects how easily patients can find and evaluate the doctor.

Professional Information, Qualifications and Specialization

The profile records the doctor's specialization, qualifications and years of experience. These fields can be edited from the profile page and are reflected immediately in the public doctor directory once saved.

Registration Information

Professional registration details, such as a medical council or licensing reference are recorded as part of the doctor profile and may be reviewed by a hospital administrator as part of the verification process described below.

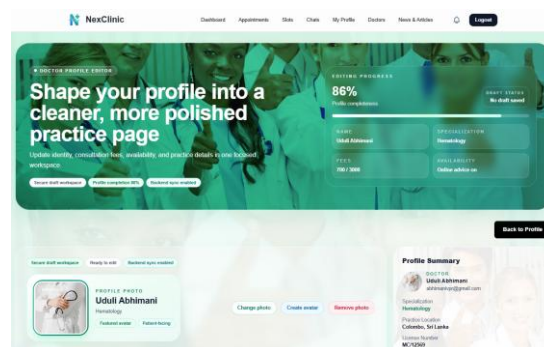


Fig. 25. Doctor profile screen

Hospital Association Requests

A doctor gains visibility to patients at a given hospital by requesting association with that hospital. The request is reviewed by a hospital administrator, who can verify or reject it, as described in Section VII.

1. Open “Hospital Associations” from the doctor dashboard.
2. Select “Request Association” and choose the hospital to request.
3. Submit the request. Its status appears as pending until a hospital administrator responds.

Hospital Verification Status

Each hospital association shows a status such as pending, verified or rejected. A doctor typically becomes visible in that hospital's directory listing, and can create slots at that hospital, only once the association is verified.

Consultation Fees

A doctor can set a consultation fee for each hospital association, since fees may reasonably differ between locations. The fee entered here is the figure shown to patients during booking.

Appointment Slot Management

Appointment slots define the specific dates and times at which a doctor is available to see patients at a given hospital. Slots can be created individually or generated in bulk from a schedule template set up by a hospital administrator, described in Section VII.

Slot Creation

1. Open “Manage Slots” from the doctor dashboard.
2. Select “Add Slot” and choose the hospital, date, start time, end time and capacity.
3. Save the slot. It becomes visible to patients searching for that doctor and hospital subject to any hospital level review that may apply.

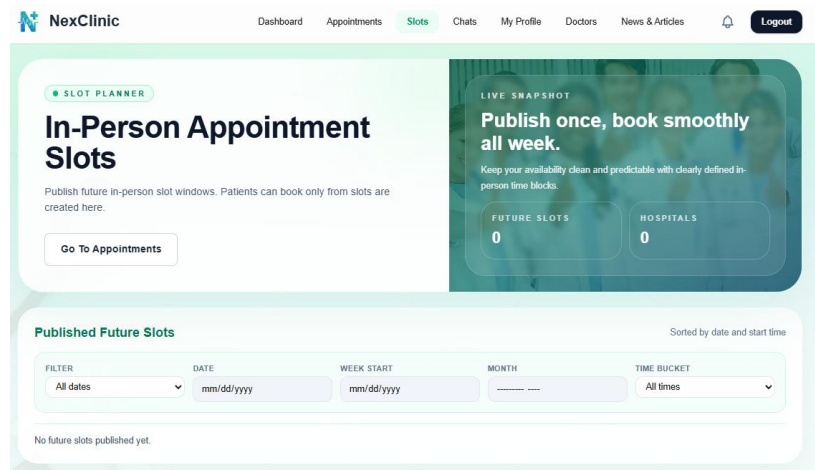


Fig. 26. Doctor slot management screen

Slot Editing and Removal

An existing slot can be edited or removed as long as it remains eligible for changes typically, while it has not yet been booked, or while any bookings against it can still be safely reassigned or notified. Removing a slot that already has a booking is handled carefully by the system to avoid silently discarding a patient’s appointment; where this situation arises, the appropriate cancellation or rescheduling workflow should be used instead.

Capacity Management

Some slots may allow more than one booking, reflecting group sessions or overlapping short consultations. The capacity value set when creating or editing a slot controls how many patients may book against it before it is shown as full.

Incoming Appointments

New bookings appear in the doctor’s appointment list with a “Pending” status until the doctor responds.

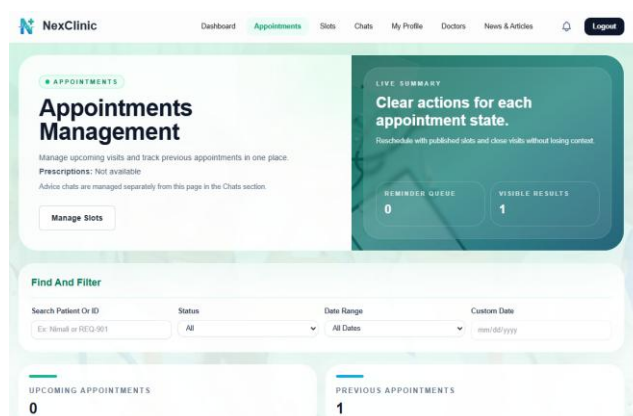


Fig. 37. Doctor appointment management screen

Doctor Action	Effect on the Appointment
Accept	Moves the appointment from Pending to Accepted, confirming that the doctor will see the patient at the requested time.
Reject	Moves the appointment to Rejected. Used when the doctor cannot honor the requested slot and no alternative is being offered.
Propose New Time	Moves the appointment to Reschedule Pending and records the doctor's suggested alternative date and time for the patient to review.
Complete	Moves an Accepted appointment to Completed once the consultation has taken place, enabling clinical-record entry.

Table 6. Doctor appointment actions and their effects

Accepting, Rejecting and Proposing a New Time

1. Open the pending appointment from the appointment list.
2. Review the patient and requested slot details.
3. Select “Accept” to confirm, “Reject” if the slot cannot be honored or “Propose New Time” to suggest an alternative.
4. If proposing a new time, select an alternative slot and submit it for the patient to review.

Completing an Appointment

1. After the consultation, open the Accepted appointment.
2. Select “Complete” once the consultation has taken place.
3. Proceed to enter clinical information as described below before or as part of completing the appointment.

Medical Record Entry

For appointments the doctor is authorized to act on, NexClinic provides a structured area for recording the outcome of a consultation.

Authorized Patient Information Access

A doctor can view a patient's relevant medical history and previously recorded information only in connection with an appointment relationship, consistent with the principle that clinical information should be visible only to those with a legitimate reason to see it.

Consultation Notes and Diagnosis

1. Open the completed or in progress appointment.
2. Select “Add Consultation Notes” and record relevant observations and the diagnosis.
3. Save the entry. It becomes part of the patient's medical history, visible to the patient and to doctors with a legitimate reason to view it.

The screenshot displays the NexClinic web application interface for entering a medical record. The top navigation bar includes links for Dashboard, Appointments, Slots, Chats, My Profile, Doctors, News & Articles, and a Logout button. The main content area shows a modal window titled "Medical Record" for patient "Abhi" on "2026-07-27" at "Test Hospital". The modal is divided into several sections: "Observations" (with a text area for symptoms, exam findings, and other notes), "Diagnosis / Assessment" (with a text area for working diagnosis, clinical assessment, or plan), "Comments" (with a text area for doctor comments, warnings, or clinical remarks), "Recommended Tests" (with a text area for lab work, scans, or referrals), "Follow-up Date" (with a date picker set to "mm/dd/yyyy"), and "Follow-up Notes" (with a text area for review instructions, red flags, or next steps). A "Context" sidebar on the right lists patient information and a "What to capture" checklist with items: "Observed symptoms and examination notes", "Assessment or diagnosis", "Prescribed medicines and usage instructions", "Recommended tests or referrals", and "Follow-up date and next-step guidance". At the bottom of the modal are "Cancel" and "Save Medical Record" buttons. Below the modal, there are tabs for "Medical Record", "Prescription", and "View Patient Profile".

Fig. 28. Medical record entry screen

Prescription Entry

1. From the same appointment, select “Add Prescription”.
2. Enter the medication, dosage and instructions.
3. Save the prescription. It becomes visible to the patient through their prescriptions and medical history view.

Fig. 29. Prescription entry screen

Clinical Document Attachments

A doctor can attach supported clinical documents to a consultation record, such as a referral letter or a scanned report, subject to the same type and size restrictions that apply to patient uploads.

Online Advice Availability

A doctor controls whether advice chat is currently offered to patients by toggling availability from the doctor dashboard. When unavailable, patients cannot start a new chat thread with that doctor, although existing threads remain visible.

Doctor Chat

Doctors use the same underlying chat workspace described for patients in Section V, with the addition of the ability to close a thread once the matter it covers has been resolved.

Closing an Advice Thread

1. Open the relevant chat thread.
2. Select “Close Thread” once the conversation has reached a natural conclusion.
3. The thread becomes read-only and remains available for later reference by both participants.

VII. Hospital Administrator Operations

Hospital Dashboard

The hospital administrator dashboard summarizes pending doctor verification requests, upcoming appointments across the hospital and recent activity, giving an administrator an overview of the hospital's current state in NexClinic.

Hospital Profile

The hospital profile stores the hospital's name, address and contact details, which are shown to patients browsing the doctor directory. Keeping this information current helps patients reach the hospital directly when needed.

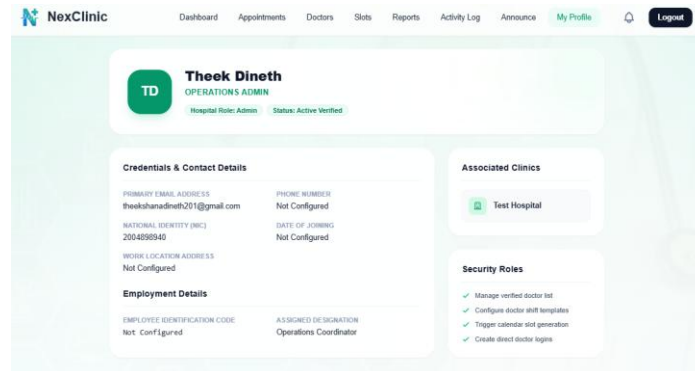


Fig. 30. Hospital profile screen

Doctor Management

Doctor Association Requests and Verification

1. Open “Doctor Requests” from the hospital dashboard.
2. Review the list of doctors who have requested association with the hospital, including their profile and registration information.
3. Select “Verify” to approve the association or “Reject” if the request should not proceed.

Doctor Fees

A hospital administrator can review and, depending on hospital policy, adjust the consultation fee associated with a doctor at that hospital, ensuring fee information shown to patients stays consistent with hospital policy.

Schedule Templates and Slot Generation

Rather than creating every appointment slot individually, a hospital administrator can define a reusable schedule template describing which days of the week and time ranges a doctor is generally available and then generate actual appointment slots from that template for a chosen period.

Creating a Schedule Template

1. Open “Schedule Templates” from the hospital dashboard.
2. Select “New Template” and define the days and time ranges that make up the template.
3. Save the template so it can be assigned to one or more doctors.

Fig. 31. Schedule template screen

Assigning Templates and Generating Slots

1. Open a saved template and select “Assign to Doctor”
2. Choose the doctor and the date range over which slots should be generated.
3. Select “Generate Slots”. NexClinic creates individual appointment slots that match the template for the chosen period.
4. Review the generated slots and adjust or remove any that need to differ from the template.

Fig. 32. Slot generation screen

Reviewing Generated Availability

Once slots have been generated, they behave exactly like slots created individually by the doctor and appear in the doctor’s own slot management view described in Section VI.

Hospital Appointment Management

A hospital administrator can view appointments across all of the hospital’s doctors, which is useful for handling exceptional situations, such as a doctor becoming unexpectedly unavailable.

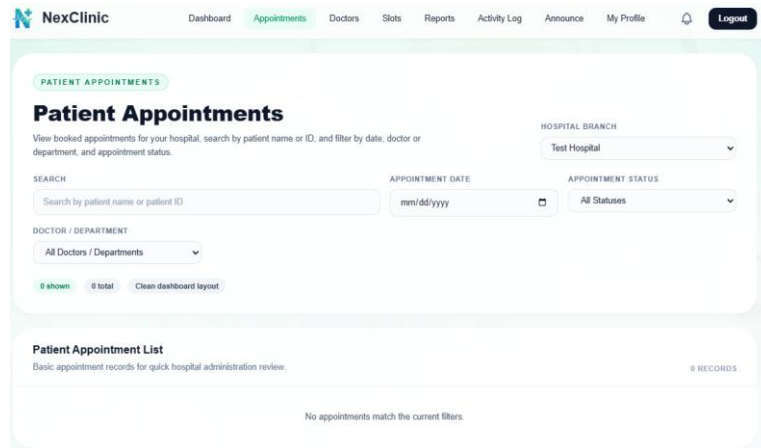


Fig. 33. Hospital appointment management screen

Appointment Cancellation

1. Open the appointment that needs to be cancelled.
2. Select “Cancel” and enter a reason for the cancellation.
3. Submit the cancellation. The patient is notified and the appointment’s status changes to “Cancelled”.

Announcements

Hospital administrators can publish announcements to keep patients and doctors informed of hospital-wide information, such as service changes or holiday schedules.

1. Open “Announcements” from the hospital dashboard.
2. Select “New Announcement” and enter a title and message.
3. Choose the target audience, such as all patients, all doctors or doctors within a particular specialization.
4. Attach a supported file if needed and publish the announcement.

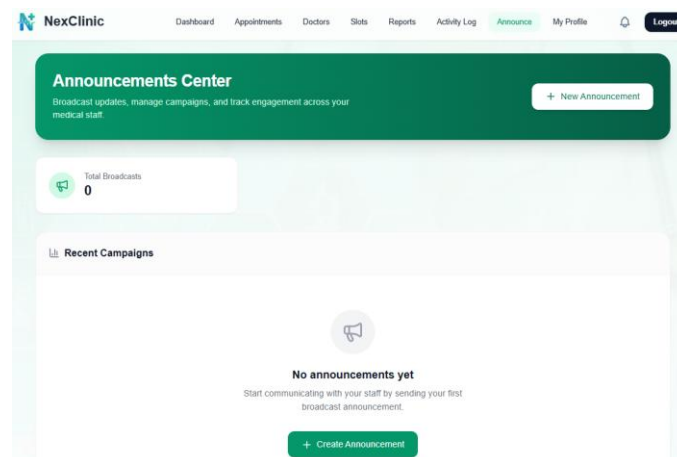


Fig. 34. Announcement screen

Published announcements are delivered to the selected audience through the notification system described in Section V and remain available in an announcement history for later reference.

Reports and Analytics

The reports and analytics area presents summary figures for the hospital, such as appointment volumes over time and doctor level statistics, helping an administrator understand how the hospital is being used.



Fig. 35. Reports and analytics screen

Activity Logs

Significant administrative actions, such as doctor verification decisions and appointment cancellations, are recorded in an activity log that a hospital administrator can review, supporting accountability and troubleshooting.

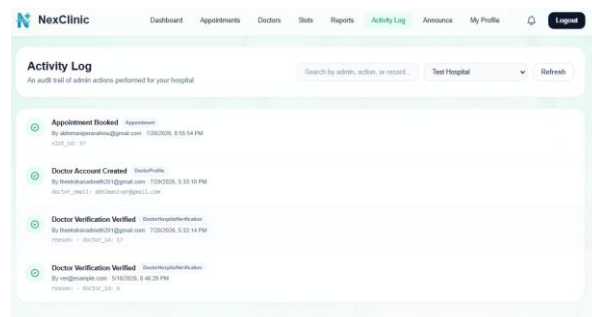


Fig. 36. Activity logs screen

VIII. System Administrator Operations

System administrator functions sit above the hospital administrator level and are concerned with the platform rather than with a single hospital. In the current implementation, a meaningful share of this work is carried out through the Django administration interface that ships with the backend framework, rather than through a dedicated system administrator workspace built specifically for NexClinic. This section describes the responsibilities involved and notes where the available tooling is administrative rather than purpose built.

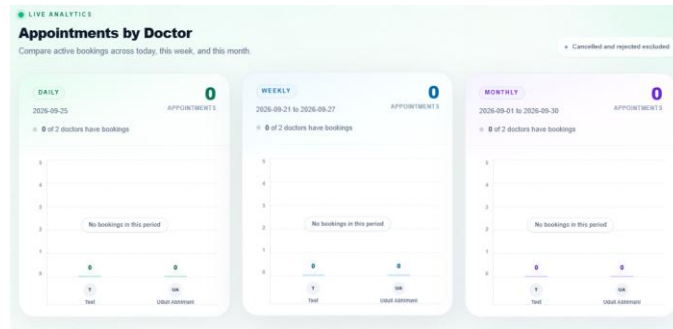


Fig. 37. System user administration screen

Administrator Access

System administrator access is granted internally rather than through public self registration, reflecting the sensitivity of the privileges involved. A person given this access should already hold or be given, a properly provisioned account with the administrator role.

User Management

A system administrator can review user accounts across all roles, correct account information where a lower-level workspace does not provide the necessary control and investigate reported problems that involve account state.

Role Management

1. Open the administrative interface and locate the account to be changed.
2. Review the account's current role and status.
3. Apply the necessary change, such as correcting a misassigned role and save.

Account Activation and Deactivation

Accounts can be deactivated to suspend access without deleting the underlying record, which is useful when an account needs to be temporarily disabled pending investigation or permanently disabled when a person leaves an organization. Reactivation restores access once the account is confirmed to be in good standing.

Administrative Data Management

Beyond individual accounts, a system administrator may need to correct or adjust reference data that supports the rest of the platform, such as fixing a malformed hospital record. This work should be performed carefully, since changes at this level can affect many dependent records.

Audit Review

Activity logs recorded throughout NexClinic, including hospital level logs described in Section VII, give a system administrator a way to reconstruct what happened in a disputed or unclear situation.

Data Retention Responsibilities

A system administrator shares responsibility, together with the wider project team, for deciding how long different categories of data, including accounts, appointments, clinical notes, chat history and activity logs, should be retained and for applying deletion or archiving consistently with that decision. NexClinic does not currently enforce a specific retention schedule on its own, any such schedule would need to be defined and applied as an operational policy.

IX. Troubleshooting

Table 7 lists problems that a NexClinic user may encounter, along with a likely cause and a recommended first action. Persistent problems that are not resolved using this table should be reported to a hospital administrator or system administrator, who can investigate further using the tools described in Section VIII and in the Developer Guide.

Problem	Likely Cause	Recommended Action
OTP does not arrive	Email delivery delay or an incorrect email address on the account.	Wait briefly, check spam or junk folders, confirm the registered email address and use "Resend Code" if needed.
OTP expired	Too much time passed between the code being issued and submitted.	Request a new code using "Resend Code" and enter it promptly.
Invalid password	The password entered does not match the stored account credentials.	Reenter the password carefully, checking for caps lock, or use "Forgot Password"
Repeated login failure	Multiple incorrect password attempts have triggered a temporary lockout.	Wait for the lockout period to end, or reset the password.
Password-reset email does not arrive	Email delivery delay or the address entered does not match a registered account.	Check spam or junk folders and confirm the email address used at registration.
Wrong workspace opens	The account was registered under a different role than expected.	Confirm which role the account was registered as and register a separate account for a different role if genuinely needed.
Doctor is not visible	The doctor's hospital association has not yet been verified or the doctor has no available slots at the searched hospital.	Try a different hospital filter or check back once verification is expected to be complete.
No appointment slots are available	The doctor has not yet created or generated slots for the desired period.	Try a different date range or contact the hospital directly for the doctor's availability.
Appointment cannot be cancelled	The appointment has moved past the stage at which cancellation is offered, such as Completed.	Contact the hospital administrator if cancellation is genuinely required after this point.

Problem	Likely Cause	Recommended Action
Rescheduling action is unavailable	The appointment is not currently in a Reschedule Pending state.	Check the current status; rescheduling response is only available while that status applies.
Chat is unavailable	The doctor has marked advice chat as unavailable or the WebSocket connection has dropped.	Confirm the doctor's advice-chat availability and refresh the page to attempt reconnection.
Notifications are delayed	A temporary delay in real time delivery or background processing.	Refresh the notification centre, the underlying event has still been recorded even if the alert is late.
Upload fails	An unsupported file type, or a network interruption during upload.	Confirm the file type is supported and retry on a stable connection.
File is too large	The file exceeds the maximum allowed upload size.	Reduce the file size, for example by compressing an image, and try again.
Session expires	The login session has reached its time limit or the token could not be refreshed.	Sign in again; unsaved form data may need to be reentered.
Dashboard data does not load	A temporary connectivity issue or backend delay.	Refresh the page, if the problem persists, report it to an administrator.

Table 7. Common problems and recommended actions

X. Privacy and Responsible Use

Protecting Personal and Healthcare Information

NexClinic stores personal and clinical information that deserves careful handling. Patients should provide only the information genuinely relevant to their care, and should review who can see a piece of information. For example, a chat message versus a formal medical record before sharing something sensitive.

Secure File Handling

Uploaded documents and attachments should be limited to what is actually needed for care or administration. Avoid uploading documents that contain information unrelated to the purpose of the upload.

Appropriate Chat Use

Advice chat is intended for care-related communication with an eligible doctor. It is not a substitute for a full consultation when an in person or scheduled examination is clinically appropriate and it should not be used to send information that has no bearing on the patient's care.

Shared Device Safety

On a shared or public device, avoid saving passwords, always sign out at the end of a session and close the browser tab afterward so that dashboard information is not left visible to the next user.

Responsible Administrative Access

Hospital and system administrators have access to information belonging to many patients and doctors. This access should be used only for legitimate administrative purposes, consistent with the least privilege principle described in the Developer Guide.

XI. Known Limitations

The following limitations apply to the current state of NexClinic and should be understood by anyone using or evaluating the system.

- The payment workflow at appointment booking is a mock or simulated gateway and does not represent a real production financial transaction.
- Online chat depends on WebSocket and network availability and messages may be delayed or fail to appear if the connection is interrupted.
- Email OTP delivery depends on SMTP or email provider configuration and may be delayed by factors outside NexClinic's control.
- Production file storage requires persistent storage, uploaded files are only as durable as the storage configuration in place.
- Some system-administration functions rely on the Django administration interface rather than a purpose-built NexClinic screen.
- Some frontend administration screens may be minimal in their current form and may be expanded in future work.
- NexClinic supports healthcare administration and communication but does not replace professional medical judgment.
- NexClinic is not an emergency service platform.
- Production use of NexClinic would require additional security, privacy, clinical, legal and regulatory review appropriate to the jurisdiction and setting of deployment.
- NexClinic does not claim compliance with HIPAA, GDPR or any other specific regulation, no such compliance has been formally assessed.

XII. Glossary

Term	Definition
API	Application Programming Interface. A defined way for one piece of software, such as the NexClinic frontend, to request data or actions from another, such as the backend.
Appointment Slot	A specific date, time and hospital at which a doctor has indicated availability for a consultation, which a patient can book.
ASGI	Asynchronous Server Gateway Interface. A server interface standard that allows a Python web application to handle asynchronous features such as WebSocket connections, in addition to regular HTTP requests.
Authentication	The process of confirming that a user is who they claim to be, typically through a password and, in NexClinic, an additional OTP step.

Term	Definition
Authorization	The process of determining what an authenticated user is permitted to see or do, based on their role and relationship to specific data.
Dashboard	The main landing page shown to a user after signing in, summarizing information relevant to their role.
Django	A Python web framework used to build the NexClinic backend.
JWT	JSON Web Token. A compact, signed format used to represent an authenticated session, allowing the backend to verify a user's identity on each request without a separate lookup.
Notification	A short message generated by NexClinic to inform a user of a relevant event, such as an appointment update or a hospital announcement.
OTP	One-Time Password. A short numeric code, valid for a limited time, sent to a user's email address to confirm identity during registration or login.
Patient Profile	The stored record of a patient's personal, contact, emergency-contact, and health-related information within NexClinic.
Prescription	A record created by a doctor specifying medication, dosage, and instructions for a patient, following a consultation.
Refresh Token	A credential used to obtain a new access token once the current one expires, without requiring the user to sign in again.
Role Based Access Control	A security approach in which what a user can see or do is determined by their assigned role, such as patient, doctor, hospital administrator or system administrator.
Schedule Template	A reusable definition of a doctor's typical weekly availability, used by a hospital administrator to generate multiple appointment slots at once.
WebSocket	A communication protocol that keeps a connection open between the browser and the server, used by NexClinic to support real-time chat and live notification delivery.

Table 8. Glossary of terms used in this manual

XIII. References

The following sources were used in the preparation of this manual, primarily to confirm terminology and general behavior for the technologies underlying NexClinic. Full bibliographic detail, including access dates, is given in the Developer Guide, which cites the same sources in the context of their technical use.

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